

MLFF CUEQ-PHASE1 Training-Only Qualification

Phase-separated e3nn source execution and pure-cuEquivariance training

2026-08-15

1. Scope

Gate: CUEQ-PHASE1

Implementation release: mdstats 0.20.189a0

Qualification schedule: FINAL-GPU1

CUEQ-PHASE1 implements the scientific-evidence and execution-separation authority for the first accelerator training experiment. It does not run or claim intermediate GPU qualification.

The frozen execution split is

$$\text{source inference / DATA6 / pseudolabels / source evaluation} = \text{e3nn}, \quad \text{TRAIN2} = \text{CuEq}_{\text{pure}}.$$

The candidate starts from the exact EXTRACT1-qualified selected-head `omat_pbe` checkpoint. The original six-head MH-1 checkpoint and exact `omat_pbe` head remain the scientific source-foundation identity.

2. CUEQ-DEP1 prerequisite

Every paired trajectory binds the same `CueqDep1RuntimeRecord.v1` digest. Positive CUEQ-PHASE1 authorization is impossible unless that runtime record passes. Missing CUDA/CuEq capability remains negative evidence and never causes e3nn fallback.

OpenEquivariance remains optional for phase 1 because MACE 0.3.16 trains with pure CuEq when CuEq training is enabled.

3. Paired protocol identity

The e3nn reference and pure-CuEq candidate must bind identical values for:

- source-foundation scientific digest;
- selected-head starting checkpoint SHA-256 and EXTRACT1 qualification digest;
- DATA8 bundle identity;
- optimizer semantics, seed, dtype, epoch budget, and update budget;
- split and deterministic data-order identities;
- objective-weight, learning-rate, stopping, and replay-policy identities; and
- validation and EVAL2 protocol identities.

The backend realization is the deliberate independent variable: reference training is `e3nn`, candidate training is `cueq_pure`.

4. Two-level qualification

The gate requires both:

1. a **short paired adaptation** with the frozen short endpoint in the 5-10 epoch interval (default 8 epochs); and
2. **at least one representative full paired authorized training trajectory**.

A passing short trajectory is an instability/divergence screen only. It cannot authorize production CuEq training by itself.

5. Scientific acceptance

Final model weights and checkpoint bytes are not required to be bit-identical. Accelerator kernels may perturb the optimization path.

The gate instead requires both sides to preserve the existing scientific authority:

- complete the frozen epoch/update budget;
- finite losses, gradients, and parameters;
- pass TRUE_DFT replay retention;
- preserve checkpoint admissibility and the existing ranking authority;
- successfully extract the selected `target_head`;
- pass EVAL2;
- pass any physical verification that is available for the reference path; and
- avoid any hard-decision disagreement between the paired runs.

Target/replay validation metrics are recorded with their paired deltas but CUEQ-PHASE1 introduces no new tolerance and does not relax any existing scientific threshold.

6. Performance evidence

Each trajectory records wall time, update throughput, peak allocated VRAM, and peak reserved VRAM. Pair evidence reports speedup as a diagnostic. Performance cannot rescue a failed scientific decision.

7. Schemas and implementation

`mdstats.training_data.cueq_phase1` defines:

- `CueqPhase1Policy.v1`;
- `CueqPhase1TrajectoryRecord.v1`;
- `CueqPhase1PairedAssessment.v1`; and
- `CueqPhase1QualificationRecord.v1`.

The gate tool is:

`tools/qualify_mlff_cueq_phase1.py`

It can build a pair record, assemble final short+full qualification evidence, or emit the current deferred fail-closed state. It deliberately does not launch GPU training.

FINAL-GPU1 preflight advances to schema `mdstats.mlff-final-gpu1.preflight.2026-08.v3` and embeds the CUEQ-PHASE1 deferred state plus the exact qualification schema expected from the final workstation campaign.

8. Authorization boundary

A passing CUEQ-PHASE1 record authorizes only:

```
source execution = e3nn
training execution = cueq_pure
```

It does **not** authorize:

- CuEq execution of the original six-head MH-1 source foundation;

- CuEq DATA6 or pseudolabel generation;
- CUEQ-PHASE2;
- a generated-default change; or
- reinterpretation of prior e3nn scientific evidence.

Those remain separate gates.

9. Development-host state

The release environment is CPU-only and lacks the CuEq core/Torch/CUDA-ops stack. CUEQ-PHASE1 therefore remains intentionally unqualified at release build time. The implementation is CPU/control-plane qualified; positive paired training evidence is deferred to the single consolidated **FINAL-GPU1** run.